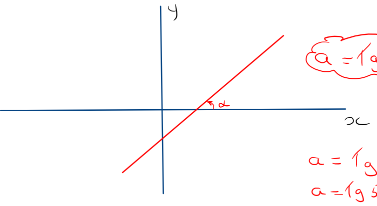
Aula 28/05/26

Função Linear

$y = ax + b \Rightarrow$ Equação Reduzida

a → Coeficiente Angular

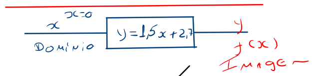
b → Coeficiente Linear

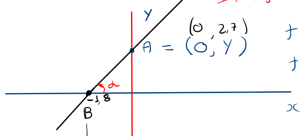


$a = Tg\,\alpha$

$a = Tg\,45°$

$a = Tg\,56{,}31°$





$f(x) = 1{,}5x + 2{,}7$

$f(0) = 1{,}5 \cdot 0 + 2{,}7$

$f(0) = 2{,}7$

$A = (0, 2{,}7)$

↳ Raiz $\Rightarrow B = (x, 0) = (-1{,}8, 0)$

$y = 1{,}5x + 2{,}7$

$0 = 1{,}5x + 2{,}7$

$1{,}5x + 2{,}7 = 0$

$1{,}5x = -2{,}7$

$x = \dfrac{-2{,}7}{1{,}5}$

$x = -1{,}8$

$a = Tg\,\alpha$

$1{,}5 = Tg\,\alpha$

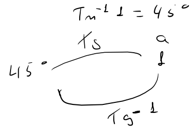
$\alpha = 56{,}31°$

$a = 1$

$a = 1 \Rightarrow a = Tg\,\alpha$

$1 = Tg\,\alpha$

$Tn^{-1}\,1 = 45°$



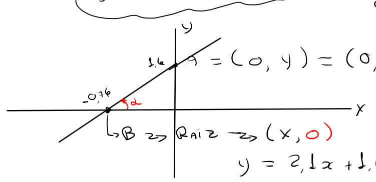
$y = 2{,}1x + 1{,}6$

$y = ax + b$

a = Coeficiente Angular

$a = Tg\,\alpha$

$2{,}1 = Tg\,\alpha \Rightarrow$



$f(x) = 2{,}1x + 1{,}6$

$f(0) = 2{,}1 \cdot 0 + 1{,}6$

$f(0) = 1{,}6$

$A = (0, 1{,}6)$

↳ B ⇒ Raiz ⇒ $(x, 0)$

$y = 2{,}1x + 1{,}6$

$0 = 2{,}1x + 1{,}6$

$2{,}1x + 1{,}6 = 0$

$2{,}1x = -1{,}6$

$x = \dfrac{-1{,}6}{2{,}1} = -0{,}76$

$B = (-0{,}76, 0)$

Calculo do Ângulo α

$a = Tg\,\alpha$

$2{,}1 = Tg\,\alpha$

$\alpha = 64{,}53°$

$y = 2x + 1 \Rightarrow$ Equação Reduzida

$0 = 2x - y + 1$

$2x - y + 1 = 0 \Rightarrow$ Equação Geral

$r: 2x - y + 1 = 0 \Rightarrow$ Equação Geral

$2x + 1 = y$

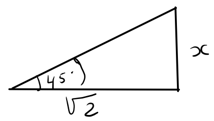
$y = 2x + 1$ | $a = 2$, $b = 1$

$r: 3x + 2y - 5 = 0$ Eq. Geral

$2y = -3x + 5$

$y = \dfrac{-3x+5}{2}$

$y = -\dfrac{3}{2}x + \dfrac{5}{2}$ $\left|\begin{array}{l} a = -\dfrac{3}{2} \\ b = \dfrac{5}{2} \end{array}\right.$



$Tg\,\alpha = \dfrac{x}{\sqrt{2}}$

$Tg\,45° = \dfrac{x}{\sqrt{2}}$

$1 = \dfrac{x}{\sqrt{2}}$

$\dfrac{x}{\sqrt{2}} = 1$

$x = \sqrt{2}$

$y = x + 2$

$y = ax + b$

$a = 1 \Rightarrow$ coef. Angular

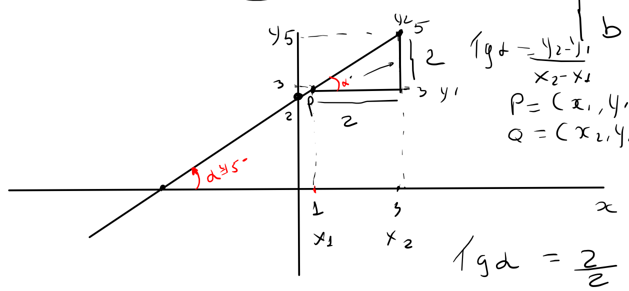
$b = 2 \Rightarrow$ coef. linear

$a = Tg\,\alpha$

$a = Tg\,\alpha$

$1 = Tg\,\alpha$

$\alpha = 45°$



$Tg\,\alpha = \dfrac{y_2 - y_1}{x_2 - x_1}$

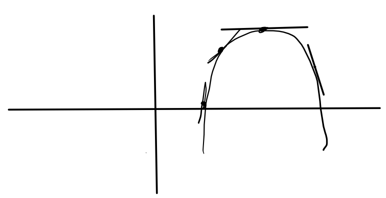
$P = (x_1, y_1)$

$Q = (x_2, y_2)$

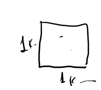
$Tg\,\alpha = \dfrac{2}{2} = 1$

$Tg\,\alpha = 1$

$\alpha = 45°$



SAGA



BIODIVERSIDADE

$\left(\dfrac{x+1}{2}\right) - 1 \cdot \left(\dfrac{x-2}{3}\right)$

$\left(\dfrac{x}{2} + \dfrac{1}{2}\right) - 1 \cdot \left(\dfrac{x}{3} - \dfrac{2}{3}\right)$

$\left(\dfrac{x}{2} + \dfrac{1}{2}\right) - \dfrac{x}{3} + \dfrac{2}{3}$

$\sqrt[5]{2^4} = (2)^{\frac{4}{5}}$

$\sqrt[2]{(x^2 - 5x + 6)^1} = (x^2 - 5x + 6)^{\frac{1}{2}}$

Notação Científica

$1\,390\,000 = 1{,}39 \times 10^{+6}$

$1 \leq N < 10$

$0{,}000539 = 5{,}39 \times 10^{-4}$

$10^{3} \Rightarrow$ Kilo

$10^{6} \Rightarrow$ Mega

$10^{9} \Rightarrow$ Giga

$10^{12} \Rightarrow$ Tera

$N = 25950000$

$N = 2{,}595 \times 10^{7}$

$2x + 1 = 0$
$2x = -1$
$x = -\frac{1}{2}$

$2x + 1 > 0$
$2x > -1$
$x > -1/2$

$-2x + 1 > 0$
$(-1)\; -2x > -1 \;.\; (-1)$
$2x < 1$
$x < \frac{1}{2}$

$4 > 2$
$-4 < -2$

$-2x + 1 = 0$
$(-1)\; -2x = -1 \;.\; (-1)$
$2x = 1$
$x = 1/2$

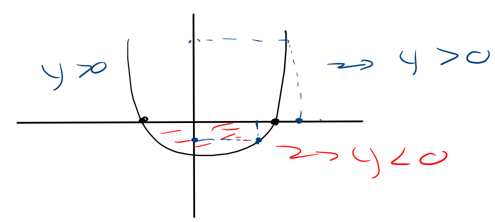
$x^2 - 5x + 6 > 0$ Inequação

y

Equação

$y > 0$

$y = x^2 - 5x + 6$



---

x → $y = \sqrt{x^2 - 4x + 3}$ → y

Domínio

$f(x)$

Imagem

If then else

Imagem

Domínio

Casos

1-) $\frac{P(x)}{Q(x)} \Rightarrow Q(x) \neq 0$

2-) $\sqrt[\text{par}]{A} \Rightarrow A \geq 0$

3-) $\sqrt[\text{impar}]{A} \Rightarrow A = \mathbb{R}$

If $(x^2 - 4x + 3) \geq 0$ then
$y = (x^2 - 4x + 3)^{1/2}$

If $x < 1$ e $x > 3$ then
$y = (x^2 - 4x + 3)^{1/2}$

$$f(x) = \frac{\sqrt{4+x}}{x-1}$$

1-) $4+x \geq 0 \Rightarrow x \geq -4$

2-) $x-1 \neq 0 \Rightarrow x \neq 1$